

one candidate θ^* out, to whichever worker is free

arrivals $(\theta_i, \rho_i, \epsilon_i, \tau_i, w_i)$

worker 1: simulate

worker 2: simulate

worker 3: simulate

$\vdots$

worker W : simulate

Evaluated history $\mathcal{H}_n$

append-only log, one row per evaluation

$\theta_i \ \rho_i \ \epsilon_i \ \tau_i \ w_i$

$\vdots \ \vdots \ \vdots \ \vdots \ \vdots$

whole
history

Propagator call (a pure function of $\mathcal{H}_n$; Alg. 1)

- 1 bandwidth $\epsilon_n = \min(\epsilon_{\text{hist}}, \epsilon_{\text{sched}})$
- 2 archive $A_n = \text{Top}_k(\mathcal{H}_n \text{ by } \rho)$
- 3 proposal $q_n = \sum_{j \in A_n} \tilde{W}_j^{(n)} K_{\Sigma_n}(\theta - \theta_j)$
- 4 candidate $\theta^* \sim q_n, w^* = \pi(\theta^*) / \bar{q}_n^{\text{on}}(\theta^*)$

W workers; none of them ever waits

No mutable sampler state.
A crashed run is recovered
by replaying its log, not
from a checkpoint.

replay the log

Reported estimator (replayed from the log, after the budget)

$$\hat{\pi}_n(f) = \frac{\sum_i W_{i,n} f(\theta_i)}{\sum_i W_{i,n}}$$

$W_{i,n} = \pi(\theta_i) K_{\epsilon_n}(\rho_i) / \bar{q}_n(\theta_i)$ for every $i \leq n$

$\bar{q}_n$: $m \leq 21$ history-reconstructed proposals q_{τ_s} + a prior floor

identical whether formed online, afterwards, or after a restart